

## 01 TEAM DETAILS

# SaakhScore

Saakh (साख) — a trader's earned creditworthiness. An explainable MSME Financial Health Card that proves it, built for IDBI Innovate 2026.

|                   |                                                     |
|-------------------|-----------------------------------------------------|
| Team name         | <b>Looper</b>                                       |
| Team leader name  | <b>Bhunesh Bansal</b>                               |
| Problem Statement | <b>Problem Statement 3 — Financial Health Score</b> |

## 02 BRIEF ABOUT THE IDEA

# Consented alternate data, turned into a decision a bank can defend.

### 01 What it is.

An explainable MSME Financial Health Card that scores businesses from consented alternate data.

### 03 Inputs.

GST filings, UPI transaction trails, Account Aggregator bank statements, EPFO records.

### 02 Who it's for.

New-to-Credit (NTC) and New-to-Bank (NTB) MSMEs that traditional bureau-based underwriting rejects outright.

### 04 Outputs.

A 0–100 score, an A–E grade, an indicative credit limit, and a plain-language improvement plan.

# Explainability and auditability, by default.

## DIFFERENTIATION

### Beyond a number

Most alternate-data scoring tools return a score. SaakhScore returns a decision a bank can defend to a regulator and a borrower can act on.

## APPROACH

### Deterministic core, generative shell

No LLM ever sits on the decision path. Independent, cross-checkable scoring engines run side by side; generative text only narrates a decision the deterministic core already made and logged.

## USP

### The regulatory gap

Of 127 AI-using regulated entities surveyed, only 15 use SHAP/LIME explainability and only 18 keep audit logs (RBI). SaakhScore ships both by default.

# What the health card actually does.

## 5-Pillar Health Score

Liquidity, Discipline, Momentum, Leverage, Digital Footprint → 0–100 score, A–E grade, indicative limit.

## Explainable ML + Shapley

A trained model with exact per-feature Shapley attribution behind every decision — no black box.

## Traditional vs. Alternate Verdict

Side-by-side comparison showing the rejected-to-approved reversal for thin-file businesses.

## Improvement Plan + Audit Log

A concrete action plan for borrowers, EWS-style monitoring signals, a counterparty-concentration limit guardrail, a deterministic underwriter next-best-action, and an append-only audit trail for every decision made.

## From consented data to an auditable decision.

01

### Data intake

AA statements, GST, UPI, EPFO



02

### Dual scoring

Rule engine + ML model, in parallel



03

### Verdict & explain

Traditional vs. alternate, Shapley reasons



04

### Memo, audit, hand-off

Memo, audit log, OCEN-aligned offer

## The rejected → approved moment.

The NTC hero (Shree Ganesh Textiles) has zero bureau history — declined instantly by a traditional process, approved on alternate data by SaakhScore.

TRADITIONAL (BUREAU-ONLY)

**Rejected**

No credit bureau history on file (New-to-Credit / New-to-Bank).

SAAKHSCORE (ALTERNATE DATA)

**Approved**

Grade A · Score 86/100 · Indicative limit ₹27,00,000.

## One service. Four inspectable control planes.

### Operational Edge

Request IDs, body ceilings, readiness probes, security headers, rate controls, and a non-root container.

### Decision + Evidence

Five transparent pillars, calibrated PD, exact TreeSHAP attribution, policy checks, and reason codes.

### Temporal Validation

Dated outcomes, 365-day maturity, chronological development/calibration/OOT, PSI, KS, AUC, and Gini.

### Promotion + Audit

Pseudonymous hash-chain evidence and fail-closed pilot startup until model, OOT, identity, and storage pass.

# A reproducible, inspectable open stack.

|                 |                                                                                                                 |
|-----------------|-----------------------------------------------------------------------------------------------------------------|
| Backend         | Python, FastAPI, Pydantic, pytest                                                                               |
| ML layer        | Calibrated monotonic XGBoost champion with native exact TreeSHAP; calibrated logistic fallback                  |
| Frontend        | Accessible, deep-linkable HTML / CSS / JS cockpit with no build step                                            |
| Deployment & CI | Non-root Docker runtime, readiness probe, Render Blueprint, two-job GitHub Actions                              |
| Stage 2 targets | Supply IDBI sandbox records to the shipped outcome/temporal gates; add SSO/KMS and durable shared audit storage |



## **Near-zero to prototype. Scoped jointly for Stage 2.**

The prototype runs entirely on an open-source stack and free-tier hosting. Stage 2 production cost scales with transaction volume and is best scoped jointly with IDBI during the sandbox mentorship phase, once real data volume is known.

## The live decision stage.

Captured from the verified submission build: a calm NTC reversal stage beside a permanent review packet with file-divider tabs — Decision, Evidence, Model, Governance, Proof, and Sources — so nothing judges need is hidden behind a modal.

The screenshot displays the SaakhScore interface. At the top, a header bar includes the SaakhScore logo, a description of the MSME credit review process, and a 'Borrower file' dropdown menu currently showing 'Shree Ganesh Textiles (New-to-Credit)'. Below this, a grid of five borrower cards is visible: Shree Ganesh Textiles (New-to-Credit) with an A grade and ₹27,00,000; Patel Wholesale Traders with a B grade and ₹32,80,000; City Corner Retail with a D grade and ₹1,53,000; Sunrise Auto Parts with a C grade and ₹7,75,000; and Himalayan Handicrafts with a B grade. The main content area features a large heading 'Bureau rejection reversed.' followed by a paragraph explaining that Shree Ganesh Textiles (New-to-Credit) is approved for ₹27,00,000 after a cash-flow, GST, UPI, EPFO-style, and bureau signals review. Below this text are two buttons: 'Read decision packet' and 'See model evidence'. On the right side, a 'Decision packet' sidebar is open, showing a 'Five-pillar health ledger' with three metrics: Liquidity (16/20), Discipline (17/20), and Momentum (14/20), each represented by a horizontal bar chart.

| Borrower                              | Grade | Amount     |
|---------------------------------------|-------|------------|
| Shree Ganesh Textiles (New-to-Credit) | A     | ₹27,00,000 |
| Patel Wholesale Traders               | B     | ₹32,80,000 |
| City Corner Retail                    | D     | ₹1,53,000  |
| Sunrise Auto Parts                    | C     | ₹7,75,000  |
| Himalayan Handicrafts                 | B     |            |

## Bureau rejection reversed.

Shree Ganesh Textiles (New-to-Credit) is approved for ₹27,00,000 after consented cash-flow, GST, UPI, EPFO-style, and bureau signals are reviewed. Reasons, model evidence, fairness checks, and proof sit in the review packet alongside.

[Read decision packet](#) [See model evidence](#)

### Decision packet

#### Five-pillar health ledger

| Metric     | Score |
|------------|-------|
| Liquidity  | 16/20 |
| Discipline | 17/20 |
| Momentum   | 14/20 |

Model evidence tab: real-SBA benchmark splits with bootstrap intervals, DeLong-tested baselines, artifact integrity, and fairness slices.

Governance packet: audit count, validation metrics, fail-closed pilot gates, pilot KPIs, fairness slices, and controls.

Proof tab: truth boundary, judging rubric, competitor gap map, runbook, and API catalog.

## Verified, not just claimed.

**167/167**

Automated tests passing

**0.962**

Real-SBA out-of-time AUC

**0.926**

Recession-stress AUC

**418k**

Real loans, real charge-offs

The v2 benchmark (GET /model/sme-benchmark) is selected across 418,947 resolved real SBA 7(a) small-business loans at natural base rates (197,716-loan train split) and holds AUC 0.9623 / KS 0.82 on a genuinely later-in-time window (FY2017-19) and 0.9255 through a 257k-loan recession stress cohort — real outcomes, real time axis, with the whole experiment registry committed. The consumer-proxy holdout (AUC 0.7497 on 4,500 rows) remains as the conduct-pillar baseline. US proxy domain, disclosed; not an IDBI calibration.

# Stage 2 gates are live. Bank data is next.

## Phase 1

### Sandbox data intake

Connect authenticated feeds, profile distributions, and validate dated outcomes through /sandbox/pilot-readiness.

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## Phase 2

### IDBI-calibrated champion

Retrain on mature chronological cohorts and promote only when the shipped deployment gate confirms true OOT evidence.

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## Phase 3

### Fairness & monitoring

Run the implemented fairness gap, PSI, and reason-code stability checks on production-scale data.

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## Phase 4

### Bedrock-powered memos

Enable the optional AWS Bedrock provider in the pilot environment, with deterministic fallback.

## 13 LINKS

# Where to find everything.

→ GitHub Public Repository      <https://github.com/bansalbhunesh/id>

→ Animated Walkthrough      [github.com/bansalbhunesh/id/blob/main/docs/demo.gif](https://github.com/bansalbhunesh/id/blob/main/docs/demo.gif)

→ Final Product Link      <https://id-ysm9.onrender.com>

→ Demo Video (YouTube)      <https://youtu.be/0I5BY6DKB1E>